Analyzing Our Impact on the Carbon Cycle Through Ordinary Differential Equations

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Abstract

Carbon dioxide acts as a greenhouse gas when stored in the atmosphere. The burning of fossil fuels and biomass upsets the natural balance of the carbon cycle by releasing long-ago stored carbon into the air, tipping the equilibrium towards the atmospheric storage. A model consisting of a system of ordinary differential equations allows for examination of the dynamics of the carbon cycle and how humans have impacted it. The results show that no 'tipping point' point exists, past which human involvement may push the system far enough that it could not recover. However, fossil fuel use still spikes atmospheric CO_2 to dangerous levels, and energy from biomass raises the steady state value of atmospheric CO_2. The paper concludes that human involvement tips the carbon cycle towards the atmospheric storage, contributing to the greenhouse effect and warming the planet. Furthermore, biomass cannot truly be considered an environmentally-friendly solution because it too contributes heavily to atmospheric carbon. Lastly, reforestation is presented as a possible solution, which variation of the plant assimilation parameter shows to be effective in reducing atmospheric CO_2.

Keywords: Carbon Cycle, Atmospheric CO_2, Ordinary Differential Equation

1. Introduction

Carbon dioxide is a fundamental, if small, percentage of our atmosphere. It's essential for life on Earth – every form of terrestrial and aquatic organic life produces or absorbs CO_2. However, in a gaseous form, CO_2 also acts as a powerful greenhouse gas, trapping energy from the sun in Earth's atmosphere and consequently warming our environment. In accordance with the widely accepted paradigm from the United Nations Intergovernmental Panel on Climate Change (IPCC), increases in CO_2 levels in our atmosphere precede parallel increases in temperature on Earth's surface [1].

This paper will examine the ever-present cycle of carbon through flows and storages found in nature, which exists in near-equilibrium without human intervention. Any carbon released into the atmosphere is consequently absorbed, and any carbon absorbed is eventually released. When human energy usage enters the equation, however, this cycle changes drastically. Unlike natural carbon usage, the human component can upset the balance of the carbon cycle and increase or lower atmospheric carbon in detrimental ways.

This project will attempt to answer the following question: **How do human carbon emissions change the behavior of the carbon cycle?** To do so, the following sub questions will be examined:

1. How does the carbon cycle function without human involvement?
2. How does burning biomass for energy compare to burning fossil fuels?
3. In what ways can we lessen the impact of our energy needs?

Using the differential model proposed by E. Eriksson and P. Welander in 1956 [2], the flow of carbon through the earth-atmosphere cycle will be tracked and the stability behavior will be examined. Then, a term will be created to represent human carbon dioxide release and examine how human involvement upsets the natural balance. [2]

2. The Carbon Cycle in Nature

The carbon cycle describes the flows of carbon from one carbon sink to another. Carbon dioxide is absorbed by plants through photosynthesis, and broken down into oxygen and, essentially, sugar. When this stored energy is eventually used by the plant, some of the stored carbon dioxide is released back into the atmosphere, in a process called respiration. Some is also released into the ground by the roots. If the plant is eaten before the energy is used, the carbon dioxide is transferred into the animal who consumes it and is released back into the atmosphere through animal respiration. Carbon dioxide is also introduced into the soil through the decomposition of leaf litter and dead

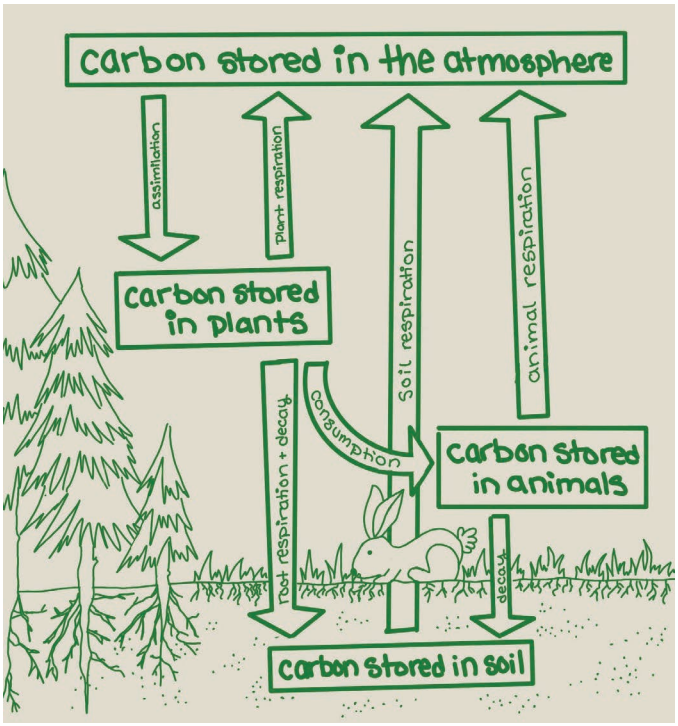


Figure 1: A simplistic representation of the earth-atmosphere carbon cycle.

organisms. The soil also respires, releasing the trapped CO_2 back into the atmosphere [3]. These processes are illustrated in Figure 1.

Lastly, there is an exchange in CO_2 between the ocean and the atmosphere, in which the ocean releases some carbon dioxide as well as absorbs some. In this paper, we will not examine the internal processes within the ocean, which are self-contained because ocean organisms both consume and release the CO_2 in solution in a relative state of balance. Therefore, although CO_2 levels in the ocean are a different area of concern, they do not widely affect atmospheric levels outside of the one exchange process [3].

3. Building the Model

3.1. Assumptions

The basic assumptions embedded in the creation of this diagram and the model it precedes are visualized in Figure 2 and described below [2]:

1. CO_2 is assimilated by terrestrial plants through photosynthesis.
2. CO_2 is produced by terrestrial plants through respiration.
3. Living organic material will become dead organic material.
4. Dead organic material will decay into CO_2.

This model engages with a timescale between $10 - 10^3$ years, which rules out both very rapid and very elongated processes, such as the internal processes contained in ocean water and the fossilization of organic matter [2].

3.2. Living Organic Matter

The first equation will represent the carbon stored in living plants, or x_p. The assimilation of CO_2 by terrestrial plants is not constant; it has been shown that plants assimilate carbon proportionally to the amount of carbon available in the atmosphere [2]. Let a describe the proportionality constant for this relationship, and y_a represent the amount of carbon in the atmosphere. There is more to consider here, including the mass of plants, perennials and annuals, plant growth rates, etc [2]. For the purposes of this paper, all information is contained in the constant a.

The processes that drain CO_2 from the living plant storage are death and respiration. As plants die, any CO_2 they store will transfer into the category of dead organic matter. This will be represented by the term $-d_p x_p$, where d_p is the rate of death. Plants also respire some of their CO_2 back into the atmosphere, at a rate of r_p [2]. Then, the equation for the rate of change of CO_2 in living plants is

$$\frac{dx_p}{dt} = ax_p y_a - (d_p + r_p)x_p. \tag{1}$$

3.3. Dead Organic Matter

CO_2 stored in the soil will be called x_d and interchangeably referred to as CO_2 stored by dead organic matter. The process that increases this amount of CO_2 are terrestrial plants dying, d_p [2]. The singular process considered that will drain this store of carbon is soil respiration [2]. Let this rate be called r_s. Then, the differential equation for the rate of change of CO_2 in the soil as

$$\frac{dx_d}{dt} = d_p x_p - r_s x_d. \tag{2}$$

3.4. Atmosphere

Lastly, y_a, the amount of carbon stored in the atmosphere, could be represented as

$$\frac{dy_a}{dt} = -ax_p y_a + r_p x_p + r_s x_d.$$

However, it is much simpler to introduce a constant T to represent the total amount of CO_2 in the system, so that

$$T = x_p + x_a + x_d + y_s + y_a,$$

or

$$y_a = T - x_p - x_a - x_d - y_s.$$

3.5. The Final System of Equations

Then, the system of equations used to describe the diagrams shown in Figures 1 and 2 is:

$$\begin{aligned} \frac{dx_p}{dt} &= ax_p y_a - (d_p + r_p)x_p \\ \frac{dx_d}{dt} &= d_p x_p - r_s x_d \end{aligned} \tag{3}$$

$$y_a = T - x_p - x_d. \tag{4}$$

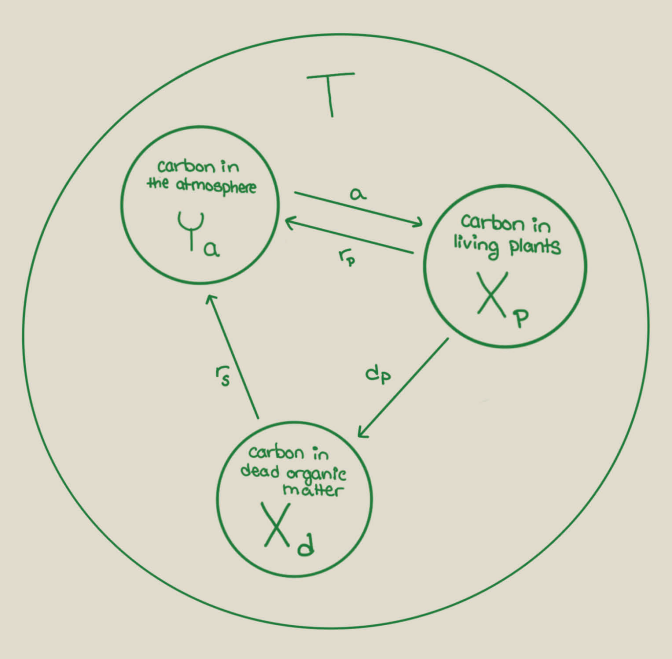


Figure 2: A schematic diagram of the system of differential equations. The model divides the total carbon, T, into three compartments representing the atmosphere, Ya, living plants, Xp, and dead organic matter, Xd.

4. The Earth-Atmosphere System

4.1. Rewriting the Equations

Eq. (4) is used to reduce to the following system of two equations on two variables, where $h = T - \frac{r_p}{a} - \frac{d_p}{a}$:

$$\frac{dx_p}{dt} = -ax_p^2 + ahx_p - ax_p x_d \tag{5}$$

$$\frac{dx_d}{dt} = d_p x_p - r_s x_d. \tag{6}$$

This will make the following analysis more easily completed and understood.

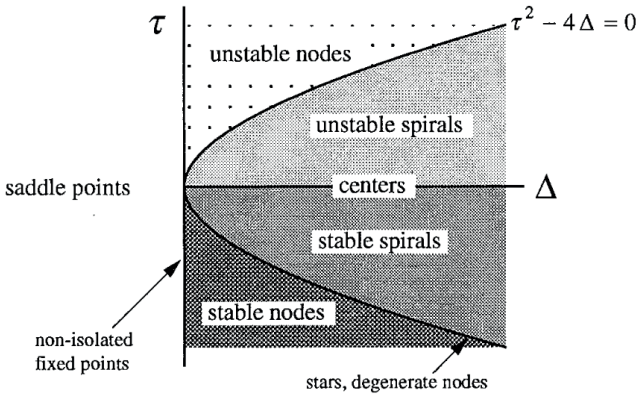


Figure 5.2.8

Figure 3: A diagram of trace-determinant stability analysis. From Nonlinear Dynamics and Chaos by S.H. Strogatz [4]

4.2. Critical Points and Stability Analysis

To examine the long term behavior of this system, the critical points and their stability are determined. The critical points of this system are $(0,0)$ and $\left(\frac{hr_s}{r_s+d_p}, \frac{hd_p}{r_s+d_p}\right)$ (Appendix A).

For the trivial critical point $(0,0)$, Appendix A shows that the eigenvalues for the Jacobian matrix are $\lambda = ah$ and $\lambda = -r_s$, where r_s represents a rate of carbon being released from the soil as dead plant matter decays. This is a positive constant. As defined above, $ah = aT - r_p - d_p$. In context, a, T, r_p, and d_p are all positive. T is defined as the total carbon contained in the atmosphere. Therefore, it is safe to assume that aT is much greater than $r_p + d_p$. Assuming that both h and r_s are real, there is one positive real and one negative real eigenvalue, and $(0,0)$ is an unstable saddle point.

For the nontrivial critical point $\left(\frac{hr_s}{r_s+d_p}, \frac{hd_p}{r_s+d_p}\right)$, Appendix A shows that the linearized system valid near the fixed point is

$$\frac{d}{dt}\begin{pmatrix} u \\ w \end{pmatrix} = \begin{bmatrix} -\frac{ahr_s}{r_s+d_p} & -\frac{ahr_s}{r_s+d_p} \\ d_p & -r_s \end{bmatrix} \begin{pmatrix} u \\ w \end{pmatrix}$$

for $u = x_p - \frac{hr_s}{r_s+d_p}$ and $w = x_d - \frac{hd_p}{r_s+d_p}$. The trace of this matrix is

$$\tau = -\frac{ahr_s}{r_s + d_p} - r_s = -\frac{(aT - r_s - d_p)r_s}{r_s + d_p} - r_s. \tag{7}$$

The determinant of this matrix is

$$\Delta = ahr_s = (aT - r_s - d_p)r_s. \tag{8}$$

As seen in Figure 3, a critical point is stable if the trace is negative and the determinant is positive. For the critical point $\left(\frac{hr_s}{r_s+d_p}, \frac{hd_p}{r_s+d_p}\right)$, Eq. (7) shows that $\tau < 0$, since $-\frac{ahr_s}{r_s+d_p}$ and $-r_s$ are negative values. Eq. (8) shows that $\Delta > 0$, since a, h, and r_s have all been previously defined as positive. Therefore, this critical point must be stable. The question remains if the point is a stable node or a stable spiral.

Thus, the carbon cycle is a stable system. Each of the carbon sinks releases and absorbs carbon in balanced amounts, resulting in long-term behavior where the amount of carbon stored in each sink stays consistent. This natural equilibrium is a well-documented phenomenon; although the levels of CO_2 in each storage change over months as plants grow and decay, yearly, the system maintains stability [3].

Further, humans cannot change the stability of the carbon cycle as represented by this model. Because all parameters must remain positive, no matter how the values are varied, $\tau < 0$ and $\Delta > 0$, and thus the system is stable. As the model defines it, the system has a total, constant amount of carbon present. Humans cannot change this amount, and therefore any CO_2 released into the atmosphere must be taken from one of the other sinks. This will have the same effect as varying a parameter, and cannot change the stability of the system. Therefore, no 'tipping point' exists beyond which human damage to the carbon cycle is irreparable. However, this does not mean that human involvement could not have disastrous effects.

4.3. Setting Default Parameters

The nontrivial critical point is expected to be a stable spiral, as CO_2 is constantly flowing from one storage to the next. Thus, the amount of carbon in each storage is constantly fluctuating, although it must approach a steady state at the fixed point. This implies spiral behavior.

As Figure 3 shows, the critical point $\left(\frac{hr_s}{r_s+d_p}, \frac{hd_p}{r_s+d_p}\right)$ will be a stable spiral if $\tau^2 > 4\Delta$, or

$$\frac{5a^2h^2r_s}{(r_s+d_p)^2} + r_s > 4ah. \tag{9}$$

The parameters used to numerically examine the behavior of the carbon cycle are, beyond the need for realistic spiral behavior, largely inconsequential. It is not necessary for their values to be life-accurate because the goal of this paper is to determine what *changes* when human involvement is added. The comparison would stand with any set of parameters that aligned with a stable spiral point.

Therefore, r_p, d_p, and r_s were varied to determine what values would result in the most severe spiral behavior, desirable because it allows changes to more easily be seen when more terms are added. Figures 4, 5, and 6 illustrate this process, showing that higher values of d_p increase spiral behavior in both x_p and x_d, while lower values of r_p and r_s do the same.

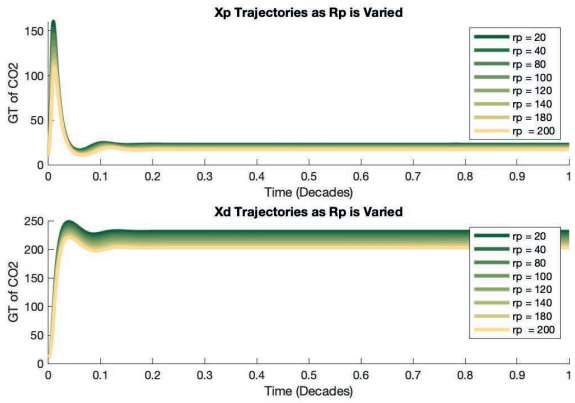


Figure 4: Results of varying r_p on the trajectories of x_p and x_d plotted against time.

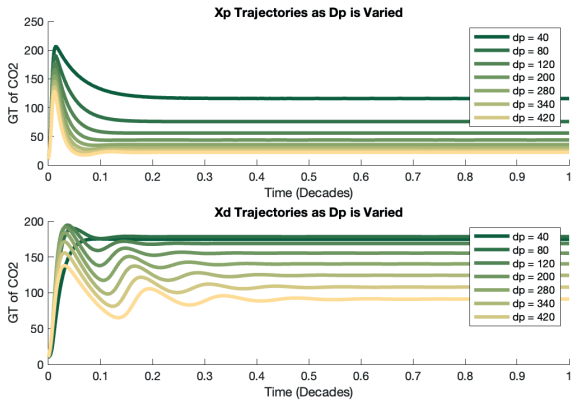


Figure 5: Results of varying d_p on the trajectories of x_p and x_d plotted against time.

Also taken into consideration was Figure 7, which shows the trajectories of x_p and x_d plotted against one another. For values of any parameter, too high or low of a value would result in the spiral deforming along the asymptotes of the nulclines. Through this process, the parameter values shown in Table 1 were chosen.

Figure 7 depicts the spiral trajectory of the system in the $x_p - x_d$ plane. The nulclines and the nontrivial critical point, $(17, 174)$, are included on the plots. The first subplot was created with initial condition $[10, 10]$, whereas subplot two used $[100, 200]$. Evidenced by Figure 7, the initial conditions of the system are inconsequential. Regardless of starting position, x_p and x_d will spiral into the critical point.

5. The Human Effect

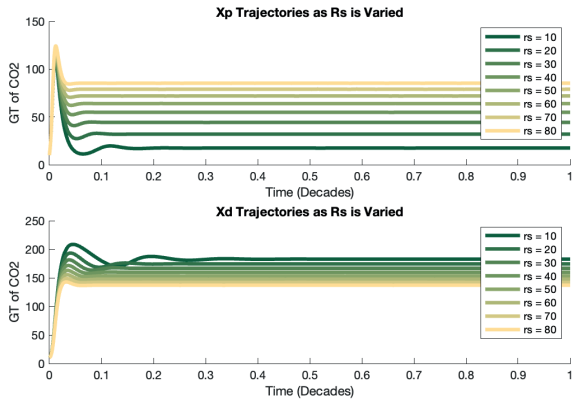


Figure 6: Results of varying r_s on the trajectories of x_p and x_d plotted against time.

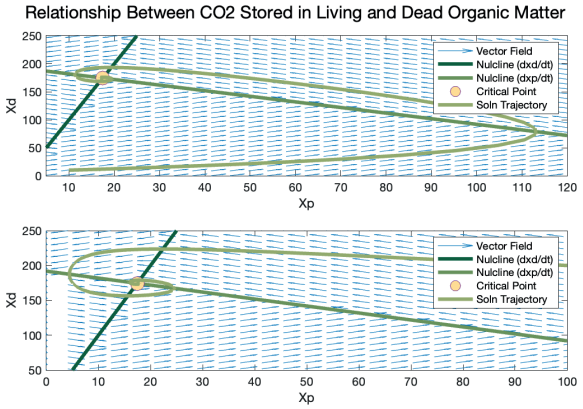


Figure 7: Direction field created using parameters from Table 1. Shows nulclines, the non-trivial critical point, and the trajectory of the solution. Subplot A uses initial condition (10, 10), and Subplot B uses initial condition (100, 200).

Param	Description	Value
a	Assimilation, rate of absorption of atmospheric CO_2 into plants.	2.25 GT per year per kg plant matter
r_p	Respiration of plants, rate of release of CO_2 trapped by plants back into the atmosphere.	150 GT per year
d_p	Death and decomposition of plants, rate of transfer of CO_2 trapped by plants into the soil.	100 GT per year
r_s	Respiration of soil, rate of release of CO_2 trapped by the soil back into the atmosphere.	10 GT per year
T	Total amount of CO_2 in the carbon cycle.	303 GT
h	$T - \frac{r_p}{a} - \frac{d_p}{a}$, combines variables to simplify results.	192 GT

Table 1: Table detailing all important parameters introduced so far, along with a description of their role and the values assigned to them.

5.1. Contextual Understanding

Since the beginning of the 18th century, humans have burned coal, oil, gasoline, and wood to produce energy. These resources contain the carbon collected over millions of years of photosynthesis, which is then released as CO_2 back into the atmosphere in a few hundred years [5]. This has contributed to an increase of approximately 150% in atmospheric carbon dioxide from 1750 [6], as illustrated in Figure 8. Figure 8 showcases data measured by NOAA at the Mauna Loa Observatory, in Hawaii [6]. This corresponds with the much steeper exponential growth of global CO_2 emissions, as pictured in Figure 9 from Our World in Data [7].

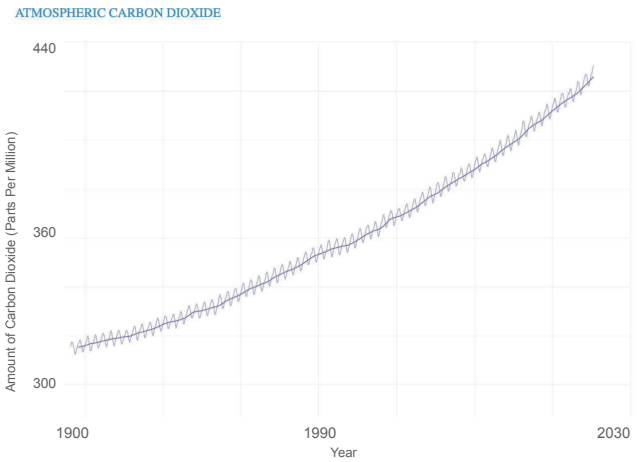


Figure 8: Plot of atmospheric CO_2 measured from the Mauna Loa Observatory, Hawaii, starting in 1958 and updated monthly. Darker blue line represents the averaged data; fluctuations of data throughout the year due to annual plant growth and decay. Modified from NOAA [5].

To represent the increase in atmospheric carbon, a term is added to $\frac{dy_a}{dt}$ governed by parameter b, representing the rate at which human activity releases CO_2 into the atmosphere. The total amount of CO_2 involved in the system, T, must remain constant. Therefore, the carbon added to y_a must come from either x_p or x_d. If the carbon was taken from x_d, this would be representative of burning coal, oil,

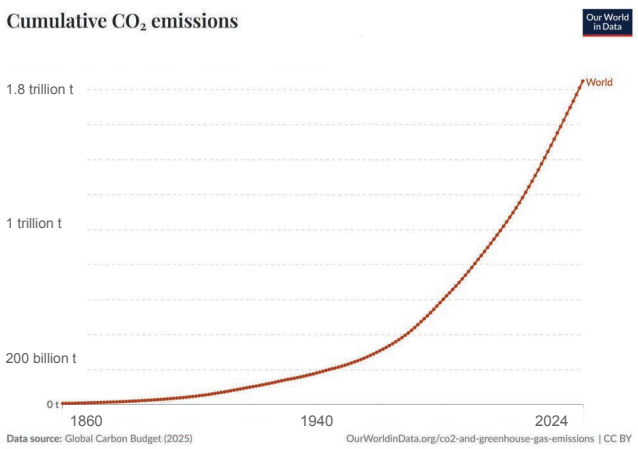


Figure 9: Cumulative global carbon emissions from 1860 to present day. Modified from Our World in Data [7].

or gas (fossil fuels) taken from within the soil. The second case, where the carbon is released from x_p, would create a representation of burning wood and other organic matter as fuel. This method of energy production, called biomass energy, is perhaps less common in modern times, but is often referred to as more 'environmentally friendly', and was even promoted by the European Union Renewable Energy Directive in 2009 as a sustainable alternative to fossil fuels [8]. Exploration of both cases will determine if biomass is truly the more carbon-neutral option.

5.2. Case 1: Fossil Fuels

In the first case, a second drain on x_d is introduced with parameter b, the rate of humans burning fossil fuels. This will correspond to an increase of bx_d in the equation $\frac{dy_a}{dt}$. The system of equations is

$$\frac{dx_p}{dt} = ax_py_a - (d_p + r_p)x_p \tag{10}$$

$$\frac{dx_d}{dt} = d_px_p - r_sx_d - bx_d \tag{11}$$

$$\frac{dy_a}{dt} = -ax_py_a + r_px_p + r_sx_d + bx_d. \tag{12}$$

Allowing $r_s + b = q$, all of the previous analysis done in Appendix A and discussed in Section 4 is still applicable. The nontrivial critical point is $\left(\frac{hq}{q+d_p}, \frac{hd_p}{q+d_p}\right)$, and must still be stable.

Figure 10 showcases the relationship between b and the amounts of carbon stored in plants, the soil, and the atmosphere. As b grows, the amount of CO_2 found in the soil, represented by the equilibrium value of x_d, shrinks. This is an expected consequence of increasing the amount of carbon extracted from the soil, which is what b represents.

Param	Description	Value
b	Fossil fuels removed from the soil and burned, releasing long-ago trapped CO_2 from the soil.	Varying
q	$r_s + b$, combines variables to simplify results.	Varying

Table 2: Table detailing all new parameters introduced in Subsection 5.2, along with a description of their role.

Also expected is that the equilibrium value of x_p increases as plants assimilate more CO_2 when more is present in the atmosphere.

Interestingly, however, the steady state value atmospheric CO_2 does not change. This is a recorded phenomenon: because plants are able to absorb more CO_2 through assimilation when there are higher levels available in the atmosphere, they drastically help to slow down the increase of atmospheric CO_2 as human emissions rise. Figure 10 shows a variation of b up to 500 GT per year, over 10 times the rate at which fossil fuels are currently burned. Still, the equilibrium value of y_a remains constant.

Mathematically, $y_a^* = T - x_p^* - x_d^*$ where the $*$ indicates a fixed point value. This is $y_a^* = T - \frac{hr_s+hd_p}{r_s+d_p} = T - h = \frac{r_p+d_p}{a}$. Because this value does not depend on r_s, variation of b has no effect on it's value.

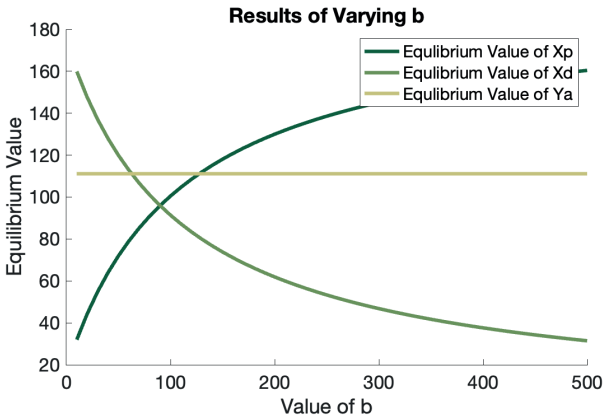


Figure 10: Plot representing the relationship between different storages of CO_2 and the value of b in Case 1: Fossil Fuels. Equilibrium values of x_p, x_d, and y_a plotted against varying values of b. Created using the parameters described in Table 1.

This implies that human use of fossil fuels has no effect on atmospheric CO_2 levels, a finding that many studies have shown this to be false. Figures 8 and 9 clearly illustrate a correlation between human CO_2 emissions and a rise in atmospheric CO_2. This unfortunate deficit in the model comes from the lack of a limit on how much CO_2

plants are able to absorb. In reality, there exists a maximum rate of carbon that can be assimilated, such that plants can only slow down the rise of atmospheric CO_2, rather than halt it entirely.

Although the equilibrium value of y_a is unchanged, the trajectory is not. Figure 11 illuminates how CO_2 levels in each storage change over time when b is varied. As b increases, a spike in atmospheric CO_2 appears in the first 0.1 years of emissions. Subplot two clarifies that this is because of a delay in plant assimilation, with the maximum level of carbon assimilated reached only after this first month.

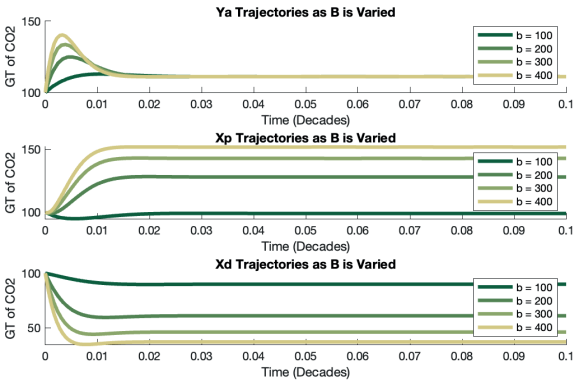


Figure 11: Trajectories of y_a, x_p, and x_d shown over time in decades as b is varied from 100 to 400. Parameter values from Table 1 are used.

If humans burn high enough levels of fossil fuels, these spikes of CO_2 can not only effect global warming, but also human health. Exposure to high levels of carbon dioxide is associated with respiratory failure, elevated blood pressure, and increased arrhythmia, or abnormal heart rhythms [9]. Thus, although the steady state value of atmospheric CO_2 is unaffected by fossil fuel emissions, some danger remains in the change in oscillating behaviors.

5.3. Case 2: Biomass

In the second case, a drain on x_p is introduced to represent carbon extracted from plant matter and burned for energy. With this new drain, again using b as the rate at which humans burn the material, the system is

$$\frac{dx_p}{dt} = ax_py_a - (d_p + r_p + b)x_p \tag{13}$$

$$\frac{dx_d}{dt} = d_px_p - r_sx_d \tag{14}$$

$$\frac{dy_a}{dt} = -ax_py_a + r_px_p + r_sx_d + bx_p. \tag{15}$$

In this case, h is updated to be $h = T - \frac{r_p}{a} - \frac{d_p}{a} - \frac{b}{a}$, resulting in the same system of equations as used in Section 4, and the same stable spiral point $\left(\frac{hq}{q+d_p}, \frac{hd_p}{q+d_p}\right)$. Table 3 shows the updated variables for this case.

Param	Description	Value
b	Burning of biomass, rate of carbon released from plant matter through burning.	Varying
h	$T - \frac{r_p}{a} - \frac{d_p}{a} - \frac{b}{a}$, combines variables to simplify results.	Varying

Table 3: Table detailing all new parameters introduced in Subsection 5.3, along with a description of their role.

The results of varying b, as shown in Figure 12, are as expected. Although the stability of the system does not change, Figure 12 shows that the location of the equilibrium point shifts drastically as b increases, with the amount stored in plants and soil both decreasing. Meanwhile, the equilibrium value of y_a grows proportionally with b.

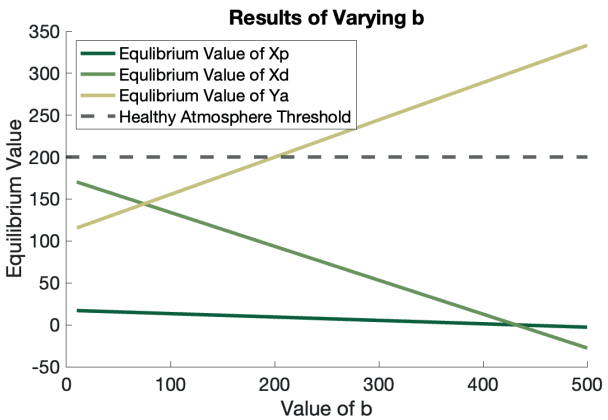


Figure 12: Plot representing the relationship between different storages of CO_2 and the value of b in Case 2: Biomass. Equilibrium values of x_p, x_d, and y_a plotted against varying values of b. Created using the parameters described in Table 1.

The more biomass that humans burn, the more atmospheric carbon is present in the atmosphere. Introducing a healthy atmospheric carbon level, set here to 200 GT simply for illustrative purposes, would allow for a maximum yearly amount of biomass used to generate energy could be found. In Figure 12, a value of $b = 200$ GT per year is the maximum before the threshold is crossed.

Beyond that level of atmospheric CO_2, the greenhouse effect would rapidly increase. Surface temperature would creep up, causing massive losses of biodiversity, food and water scarcity, and more natural disasters [10]. At a certain level, the health risks discussed in Section 5.2 would become relevant.

5.4. Fossil Fuels vs Biomass

The major difference between fossil fuels and biomass in this model are their effects on plants. Because the extraction of fossil fuels does not inherently impact the mass of available organic matter, plants are better able to offset human carbon emissions and slow the rise of atmospheric CO_2. Adversely, the inherent property of biomass energy is that plant matter must be removed from the system to be transformed into an energy source. This cripples the ability of plants to assimilate the amount of carbon needed to offset human emissions, leading to a drastic increase in atmospheric CO_2.

In both cases, the carbon cycle is tipped towards the atmospheric sink by human involvement. This is clear to see with biomass: the equilibrium values of x_p and x_d decrease, and the steady state of y_a increases, as b is increased. When considering fossil fuels, the equilibrium values greatly move towards large quantities of carbon contained in plants (Figure 11). However, also of consideration is the drastic spike of atmospheric CO_2 caused by high fossil fuel emissions before plants are able to assimilate the carbon.

6. Updating Assimilation

In Section 5.2, one limitation of the model was discovered: plant assimilation, being unlimited in capacity, can offset *any* amount of carbon dioxide released from human burning of fossil fuels. To correct this, the assimilation term can be updated to represent logistic instead of exponential growth. Then, the system of equations for the fossil fuel case is

$$\begin{aligned} \frac{dx_p}{dt} &= ax_p y_a(1 - \frac{x_p}{k}) - (d_p + r_p)x_p \\ \frac{dx_d}{dt} &= d_p x_p - r_s x_d - bx_d \\ y_a &= T - x_p - x_d, \end{aligned} \tag{16}$$

where k is the limit of CO_2 that plants are able to assimilate.

Figure 13 illustrates the updated relationship between equilibrium values and b. The equilibrium values for y_a are shown for varying k values (Appendix B). Encouragingly, Figure 13 shows that this new system of equations results in a more realistic outcome for atmospheric CO_2 levels. Further analysis can be found in the full paper. Section 7 will use the updated model with a value of $k = 100$. The model for the biomass case is analogous, with the burning term taking carbon from x_p instead, as in Section 5.3.

7. Carbon Solutions

Section 5 shows that a is a powerful parameter in this model, accurately representing the massive impact of plant

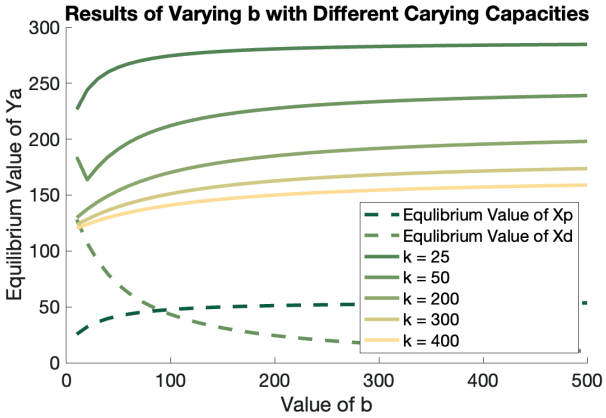


Figure 13: Plot representing the relationship between equilibrium values of y_a and b with different values of k. Equilibrium values of x_p and x_d plotted with $k = 100$. Created using the parameters described in Table 1.

assimilation on climate change. To explore this more, a is varied to represent an increase in available organic matter through reforestation or other initiatives. Note that increasing a means increasing the rate of assimilation per kg of organic material. However, the results are largely the same as increasing the mass of organic matter able to assimilate CO_2.

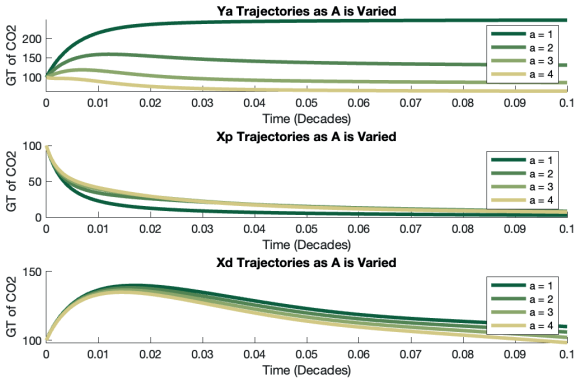


Figure 14: Trajectories of y_a, x_p, and x_d in the fossil fuel case as a is varied. Created using the parameter values described in Table 1.

Figure 14 depicts the trajectories of y_a, x_p, and x_d over 1 year while fossil fuel emissions are set to a high value, $b = 200$ GT per year. As a is varied, Figure 14 illustrates that levels of x_p increase and reach their maximum value faster, removing the assimilation lag and potentially reducing the initial spike of atmospheric CO_2 discussed in Section 5.2. x_d decreases with x_p. Most importantly, y_a is drastically reduced. When $a = 4$ GT per kg per year, atmospheric carbon reaches nearly 0 GT. This is not nec-

essarily desirable; plants and humans alike need carbon available for survival. However, Figure 14 exhibits the importance of reforestation as a climate solution.

Figure 15 shows the same data for the biomass case. Similar effects are observed. Again, increasing a decreases x_d, increases x_p, and dramatically reduces y_a.

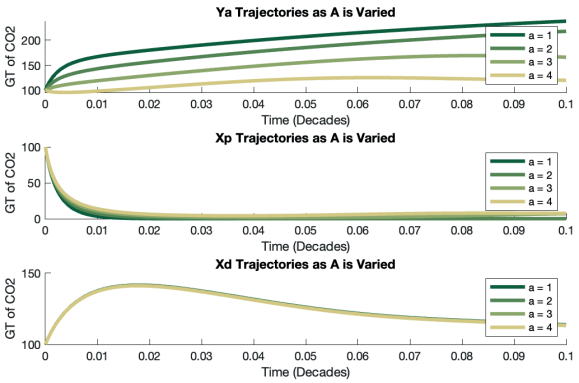


Figure 15: Trajectories of y_a, x_p, and x_d in the biomass case as a is varied. Created using the parameter values described in Table 1.

This again suggests that reforestation is a valuable option for offsetting human carbon emissions and limiting the effects human energy needs have on the natural carbon cycle. Plants are shown to be a beneficial ally in efforts to reduce atmospheric CO_2.

8. Discussion

Through modifications of Eriksson and Welander's 1956 carbon cycle model [2], the ramifications of humanity's widespread use of fossil fuels and biomass burning as energy forms were discussed. Carbon dioxide is essential for all life on earth; however, it is also a greenhouse gas in it's atmospheric form. Thus, the emphasis of this paper is on changes in atmospheric CO_2 due to human involvement.

8.1. Limitations

Although this model is useful for illustrating humanity's impact on the carbon cycle, it is limited. Firstly, the ocean is not included. In reality, the ocean constantly exchanges CO_2 with the atmosphere. The decision not to include the ocean came from Eriksson and Welander's conclusion that in the simplified form used in their extension, the ocean acts only as a dampening term to atmospheric carbon [2]. Although it would have been more accurate to include, the complications added outweighed the benefits. Similarly, animals were not included. Animals absorb CO_2 through consuming plants and release it through respiration. Their addition would have, in essence, only augmented the parameters of the organic matter storage.

Additionally, it has been shown that the respiration of both plants and soil increase as temperatures rise [11]. This is particularly concerning when the rise in global temperatures is considered, precipitating a positive feedback loop in which rising temperatures lead to an increase in CO_2, a greenhouse gas. The model cannot predict or account for this variation in plant and soil respiration.

Lastly, the model ignores many other components involved in what it means to be 'sustainable' or 'environmentally friendly'. Although the findings show that biomass performs poorly when compared to fossil fuels, there are many more factors to consider. Fossil fuels must be extracted from deep within the Earth, processed, and refined. Biomass implies some deforestation, but is overall a much less environmentally-invasive process. That being said, the findings of this paper are still applicable in indicating that biomass is not the ideal solution that many claim it to be.

8.2. Concluding Remarks

Despite limitations, the model still holds relevance today. It was shown that the system is stable for any value of parameters, implying that humans are not facing a tipping point past which reduction of atmospheric carbon is impossible.

However, both fossil fuels and biomass energy sources present significant environmental dangers through their contributions to atmospheric CO_2 levels. Adding a fossil fuel term to the original model does not effect the equilibrium values of y_a, but does create a spike in the trajectory as plants experience an assimilation lag. Assimilation was then updated to logistic growth, which allowed the model to more accurately represent the increase in atmospheric CO_2 with the addition of fossil fuels. Changing the source of energy to biomass *does* raise the steady state value of atmospheric CO_2, proportionally to amount of biomass is burned per year.

The effect on plants caused the most difference in levels of carbon dioxide in the atmosphere. This implies that reforestation is a powerful tool for carbon sequestration to minimize the greenhouse effect; variation of a to represent additional organic material confirmed this.

This paper has shown that human involvement does not destabilize the carbon cycle, but does impact what storages hold the most CO_2. Furthermore, biomass should not be considered a 'carbon-friendly' alternative to fossil fuels, because it's reduction of organic materials corresponds with steep increases in atmospheric CO_2. Lastly, reforestation should be considered an essential part of sequestering human carbon emissions.

9. Acknowledgments

Thank you to Dr. Jennifer Crodelle, both for all your help on this paper and for everything you've taught me during my time at Middlebury. Your influence on me is immeasurable.

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Megan Balparda



For most of my life, I was indifferent towards math – towards all of my classes, really. In each, I saw my friends (fun), and in each, I had to do homework (blah). End of story.

Then, in middle school, we spent a month going over and over what an equals sign meant. I was outraged! I think anyone would be. I was so bored, and so angry. Simultaneously, I was taking a separate math class where we were learning probability. I don't like probability either. It feels made up. All this to say, I went from 'indifferent' to 'passionately against' pretty quickly.

But math rebounded for me in high school. I had a great, weird teacher in my freshman year. He asked us questions about building statues out of butter, and surrounding them with ponds full of frogs. The thing is, I don't even remember the math we used to answer that. That was never really the point – he taught us how to think.

Math is creativity. It's not algebra, geometry, or calculous; these are parts of math. Methods, maybe. But math is curiosity. It's looking at the world, and thinking, can I predict what happens next? Can I describe what I'm seeing through numbers? It's problem solving, deductive reasoning. It's the unraveling of a mystery.

The people who know I'm a math major, but aren't mathematicians themselves, are quick to make jokes when I count on my fingers. When I tell someone I've never memorized the times tables, I get, 'aren't you getting a degree in math?'

'Yes', I usually reply. 'In math, not arithmetic.' It's true that I'm pretty comfortable with a fraction, and will do an integral for fun, but I'm just as likely to complain about a messy bit of algebra or trigonometry. Still, I haven't found anything more satisfying than finishing a math problem and knowing it just looks right.

When someone tells me they don't like math, I usually ask, 'well, do you like Sudoku?' We've been taught that math is hard, inaccessible, and boring. But in reality, math is like Sudoku. It's a mental exercise rooted in deductive organization. And I don't think I've ever met someone who doesn't enjoy a Sudoku puzzle every once in a while.